## SIMATS ENGINEERING
### ASSESSMENT TOOL 2

**COURSE NAME: CSA04 COURSE CODE: Operating Systems**

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**COURSE OUTCOME COVERED**

**CO1:** *Apply the concepts of Operating System types, structures, and system calls to demonstrate their functions in various computing environments. (BL2, BL3)*

*Assessment Tool Weightage: 35%*

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**QUESTIONS: 5** **Total Marks:25**

Annexure A

**Concept Mapping**

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***Code of Conduct:***

*I Priyadharshini .K.S (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
*I understand that any violation of academic integrity rules will result in disciplinary action.*

Priyadharshini KS
***Signature of the student:***

Annexure A

**Concept Mapping**

**Question 1 (5 Marks)**
Construct a multi-level concept map integrating:
Operating System Types → OS Structures → System Calls → OS Functions.
Your map must show cross-links (e.g., how OS structure influences system call implementation).

**Question 2 (5 Marks)**
Design a comprehensive concept map representing:
Process Management, Memory Management, File System, and I/O System.
Include interdependencies and indicate where system calls interact with each component.

**Question 3 (5 Marks)**
Create a comparative concept map for:
Batch OS, Time-Sharing OS, Distributed OS, and Real-Time OS.
Highlight differences, similarities, advantages, and limitations using linking phrases.

**Question 4 (5 Marks)**
Develop a dynamic concept map showing the execution flow:
User Program → API → System Call Interface → Kernel → Hardware → Response back to User.
Include feedback loops and control flow paths.

**Question 5 (5 Marks)**
Construct a hierarchical + functional concept map for system calls:
Process, File, Device, and Information system calls.
Also show how they relate to OS services like scheduling, memory allocation, and device handling.

**RUBRICS FOR CALCULATION**

| Concept Mapping | | | | | |
|---|---|---|---|---|---|
| Criteria | Weightage | Level 4 (Exemplary) | Level 3 (Proficient) | Level 2 (Developing) | Level 1 (Beginning) |
| Concept Identification | 25% | Identifies all key concepts from literature accurately and comprehensively | Identifies most key concepts with minor omissions | Identifies limited concepts; some are irrelevant or missing | Fails to identify essential concepts |
| Linkages | 25% | Establishes clear, meaningful, and accurate relationships between concepts | Shows correct relationships with minor gaps | Relationships are weak, unclear, or partially incorrect | No meaningful relationships established |
| Organization | 20% | Concepts are well-structured hierarchically with logical flow and grouping | Mostly organized with minor structural issues | Some organization but lacks clear hierarchy | No logical organization or structure |
| Integration of Literature | 20% | Integrates multiple studies effectively to show connections and comparisons | Shows some integration of literature | Limited integration; mostly isolated concepts | No integration of literature evidence |
| Clarity | 10% | Concept map is clear, neat, visually structured, and easy to interpret | Generally clear with minor readability issues | Clarity is reduced; difficult to interpret in parts | Poorly presented and hard to understand |

**Marks Evaluation:**

| Question No. | Concept Identification (25%) | Linkages (25%) | Organization (20%) | Integration of Literature (20%) | Clarity (10%) | Total Marks |
|---|---|---|---|---|---|---|
| Q1 | | | | | | |
| Q2 | | | | | | |
| Q3 | | | | | | |
| Q4 | | | | | | |
| Q5 | | | | | | |
| Overall Total (100) | | | | | | |

| Faculty In-charge | Course Coordinator | Head of Department |
|---|---|---|
| | | |
| Signature: ____________ | Signature: ____________ | Signature: ____________ |

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|---|---|---|
| Date: ____________ | Date: ____________ | Date: ____________ |

# CO1 AT2 - CONCEPT MAPPING

NAME : PRIYADHARSHINI·K·S

REG : 192511248

SUB : OPERATING SYSTEM

CODE : CSA 0404

1) Operating System Types, Structures, System calls & OS Functions :

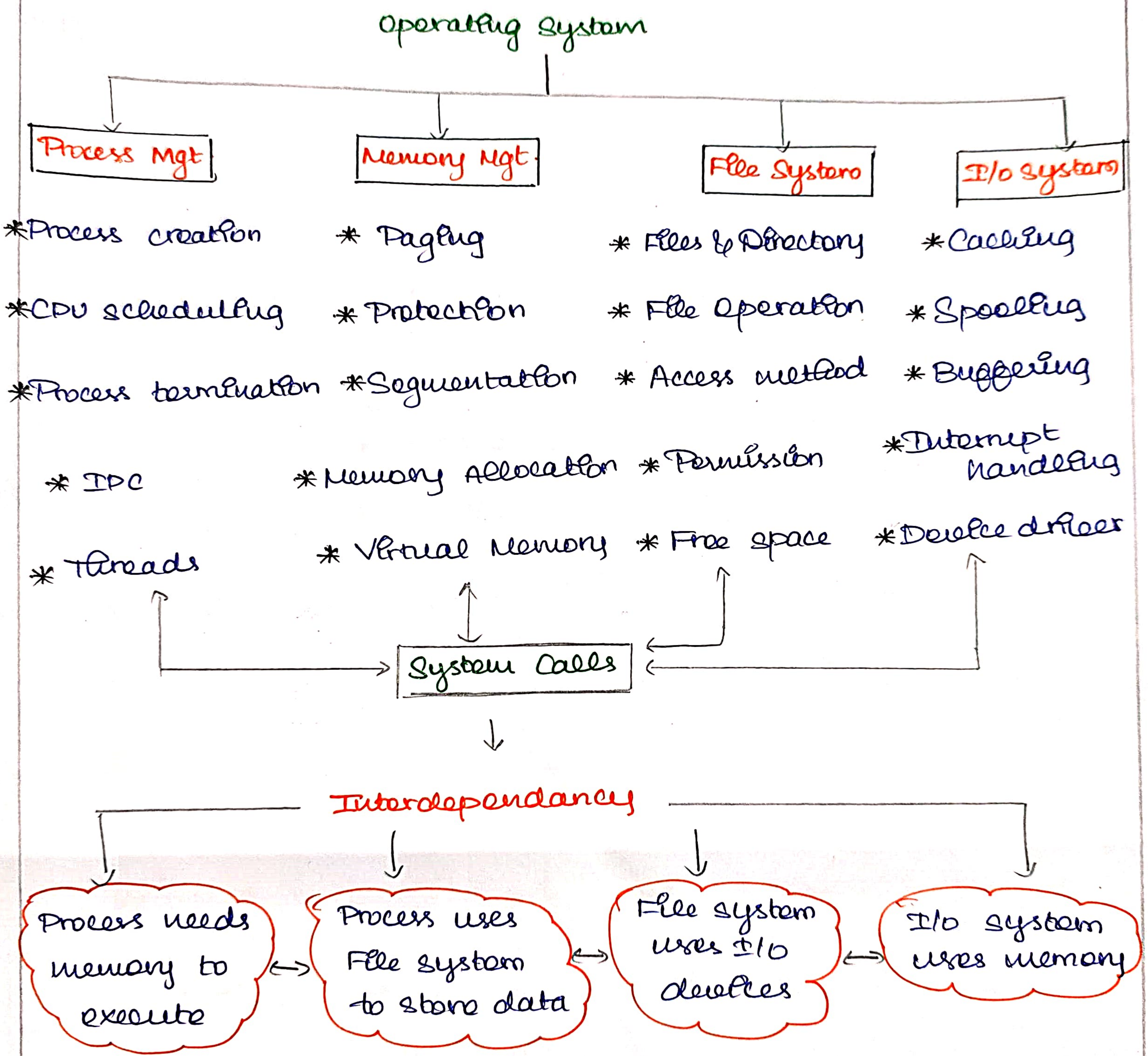
Operating System

- OS Types
  - Batch OS
  - ↓
  - Time sharing OS
  - ↓
  - Distributed OS
- OS Structures
  - Monolithic Kernel
  - ↓
  - Layered Structure
  - ↓
  - Microkernel
- System calls
  - Process calls
  - ↓
  - File calls
  - ↓
  - Memory calls
  - ↓
  - Device calls
- OS Function
  - Process
  - ↓
  - Memory
  - ↓
  - File
  - ↓
  - Device
  - ↓
  - Security

Relationships

* System calls act as interface b/w user program & OS.
* OS structures decide how components organized.
* OS Types define way resources are provided.

2) Process management, Memory Management, File system and I/O system.



3) Distributed Systems, Clustered Systems, Real-time Systems & Handheld Systems.

## Operating System Types

### Distributed System

- Multiple computer communicate over network
- Resource sharing
- Highly reliable

Limitation:

- Complex to manage

### Clustered System

- Multiple systems work together as a cluster
- Load balance
- High available

Limitation:

- High cost

### Real-time System

- Response within deadline
- Critical for time constraint
- Used in control systems

Limitation:

- Expensive

### Handheld System

- Portable
- Limited resource
- Battery operated

Limitation:

- Limited storage & power

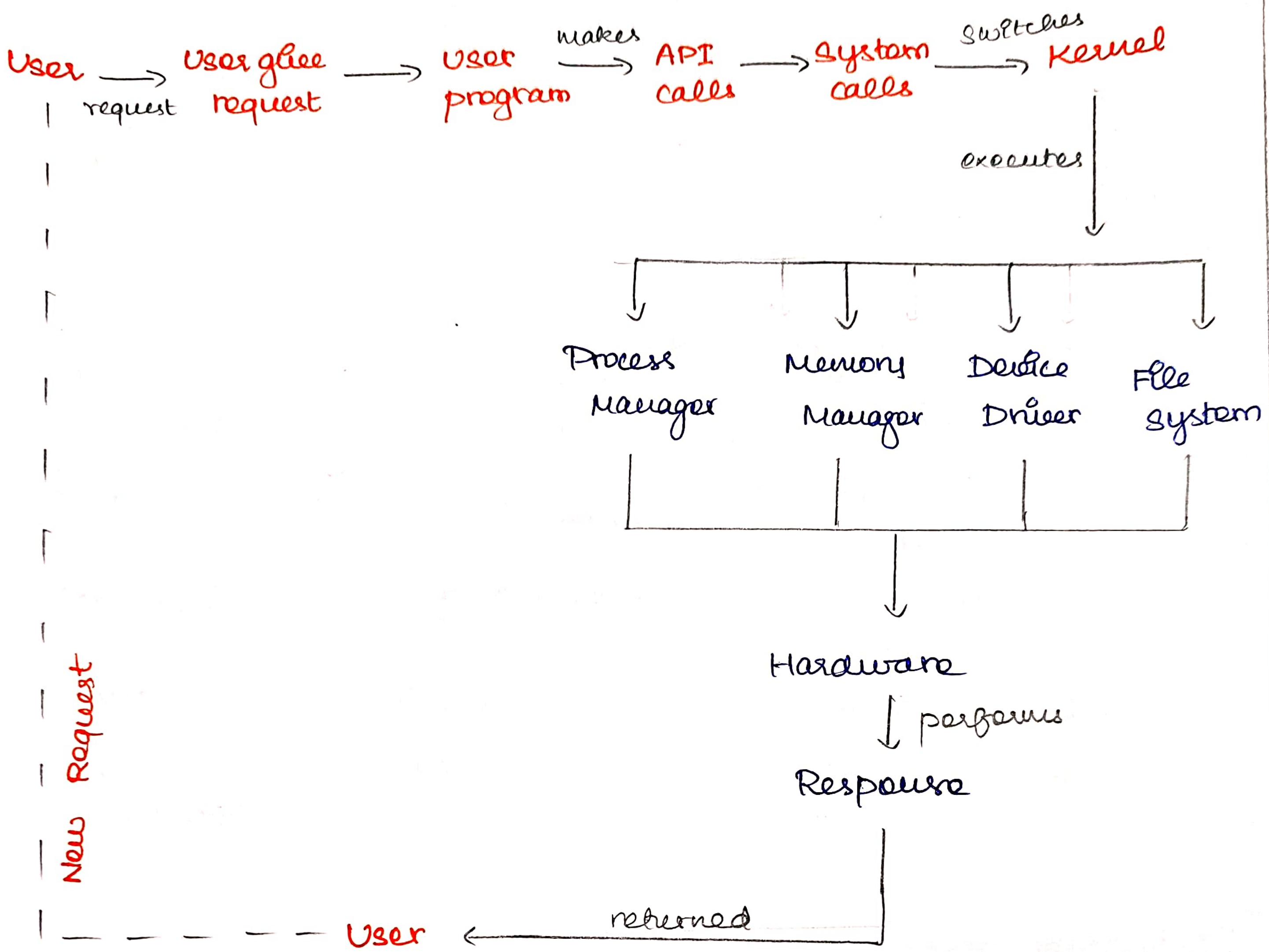
## COMPARISON

Distributed & Clustered → Focus on resource sharing & availability

Real time → Focus on deadline & quick response

Handheld → Focus on portability & limited resource

4) Execution flow of user program using System calls



## 5) System calls – Types & OS Services

### System Calls

**Process Calls**
- fork()
- exec()
- wait()
- exit()

↓ Used for process creation, termination & control

**File Calls**
- open()
- read()
- write()
- close()
- unlink()

↓ Used for file operations & Management

**Device Calls**
- ioctl()
- read()
- write()
- get device status()

↓ Used for device control & I/O operation

**Information calls**
- getpid()
- time()
- uname()
- get system info()

↓ Used to get system info & status

### OS Services

- Provide interface b/w user & hardware
- Manage resource efficiently
- Ensure security & protection

→ Process Execution